



A Guide to Misinformation Detection Data and Evaluation

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Abstract

Misinformation is a complex societal issue, and mitigating solutions are difficult to create due to data deficiencies. To address this, we have curated the largest collection of (mis)information datasets in the literature, totaling 75. From these, we evaluated the quality of 36 datasets that consist of statements or claims, as well as the 9 datasets that consist of data in purely paragraph form. We assess these datasets to identify those with solid foundations for empirical work and those with flaws that could result in misleading and non-generalizable results, such as spurious correlations, or examples that are ambiguous or otherwise impossible to assess for veracity. We find the latter issue is particularly severe and affects most datasets in the literature. We further provide state-of-the-art baselines on all these datasets, but show that regardless of label quality, categorical labels may no longer give an accurate evaluation of detection model performance. Finally, we propose and highlight Evaluation Quality Assurance (EQA) as a tool to guide the field toward systemic solutions rather than inadvertently propagating issues in evaluation. Overall, this guide aims to provide a roadmap for higher quality data and better grounded evaluations, ultimately improving research in misinformation detection. All datasets and other artifacts are available at misinfo-datasets.complexdatalab.com. The extended paper, including the appendices, can be accessed via arXiv.

- Extended paper → arxiv.org/abs/2411.05060

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CCS Concepts

- **Computing methodologies** → **Natural language processing**;
- **Applied computing** → *Law, social and behavioral sciences*.

Keywords

Misinformation Detection, Datasets, Evaluation, Survey, LLM

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1 Introduction

Misinformation is a pressing concern for society, already causing significant negative impacts and posing even greater risks with the advent of generative AI [25]. Extensive research has been devoted to this problem, yet it remains unresolved. There has been considerable recent progress in methods, especially leveraging LLMs to detect false information at scale [7]. However, to fuel further progress, we also need strong and reliable data.

Multiple studies have identified data availability, and especially data quality, as a barrier for reliable misinformation detection. To begin, obtaining high quality veracity labels is challenging and time-consuming, even for experts [30]. Shortcuts, though, can cause severe spurious correlations [18, 27], and even with high quality labels there can be issues with ambiguity of input texts [19]. While several surveys have mapped the methodological landscape in this domain [7, 16, 21, 29], the analysis of dataset quality remains either limited in scale [18, 19, 27] or lacking in depth.

To overcome this problem, we present the largest-scale survey in the literature to date, curating 75 datasets with comprehensive descriptive analyses and categorizations. This is nearly three times as many as other dataset-focused surveys like [12], and many times more than general surveys like those of [3, 16, 21, 29]. We provide

a summary of each dataset, along with key information like topic, size, modality, languages, geographic region, and time period.

We further focus on 36 datasets that include claims and 9 datasets that include paragraphs, and evaluate their quality in depth. First, we examine two types of potential spurious correlations that could lead to predictions based on invalid, non-generalizable signals. In particular, we start by looking at keyword based correlations before following up with temporal correlations. Both of these can serve as proxies for topics, events, and other correlated information. We then assess whether the examples in these datasets are actually feasible to assess for veracity at all. We find that most datasets contain substantial ambiguities and other issues, such that over half of the claims data may be infeasible for methods without evidence retrieval, with a large portion remaining infeasible even with retrieval. These quality assessments offer both immediately practical insights for selecting the most reliable datasets, and significant implications for directions in the field of misinformation research overall.

Our work also addresses challenges of model evaluation after researchers complete the dataset selection phase. We first present a unified formatting and labeling schema for all 36 claims datasets and 9 paragraph datasets. Next, we establish state-of-the-art baselines using GPT-4 with and without web search to collect evidence for veracity. Following this analysis, we find that standard evaluation metrics like accuracy and F1, when computed simply in relation to ground truth labels, are no longer sufficient to evaluate leading generative methods for misinformation detection and could lead to invalid conclusions. To address this challenge, we present and test an alternative evaluation approach, emphasizing the need for future research in this domain.

Finally, our findings come together in our recommendation for a new practice of **Evaluation Quality Assurance (EQA)**, which offers a tool to improve data selection and evaluation methodologies. In summary, we present a guide to misinformation datasets, including:

- The largest scale collection of misinformation datasets, **CDL-MD**, with a unified labeling schema, made easily accessible through a HuggingFace Repository; along with baseline performances on these datasets.
- An essential toolkit, called **CDL-DQA**, for quality evaluation of misinformation datasets to analyze spurious correlations as well as the feasibility of the datapoints. Applying this toolkit to CDL-MD reveals numerous quality issues (summarized in Table 1), such as how the majority of claims datasets contain numerous examples whose veracity may be impossible to evaluate.
- Recommended practices of **EQA**, for research proposing misinformation detection methods. Informed by our findings of severe quality issues in both data and evaluation procedures, such as how simple metrics like accuracy and F1 may be obsolete in this domain, we propose that assurance of evaluation quality is an indispensable component for future research here.

We provide links to our unified dataset collection CDL-MD on HuggingFace, our tools CDL-DQA and other code, and other outputs through our website misinfo-datasets.complexdatalab.com.

Table 1: Dataset quality assessments. A ✓ denotes a dataset that passes the evaluation criterion. The horizontal line divides claim and paragraph datasets.

✗ in Keyword or Temporal indicates supervised methods may learn significant spurious correlations of the respective type. ✗ in Feasibility indicates over a quarter of the data may be impossible to assess for veracity at all.

Dataset	Keywords	Temporal	Feasibility
AntiVax	-	✓	-
Check-COVID	✓	-	✗
ClaimsKG	✓	✗	✗
Climate-Fever	✓	-	✓
CMU-MisCOV19	-	✓	-
CoAID	✗	✓	✗
COVID-19-Rumor	✓	-	✗
Covid-19-disinformation	-	✓	-
COVID-Fact	✓	-	✗
DeFaktS	✓	✓	✗
ESOC Covid-19	-	-	✗
FakeCovid	✓	✓	✗
FaVIQ	✓	-	✓
FEVER	✓	-	✓
FEVEROUS	✓	-	✓
FibVID	✓	✗	✗
HoVer	✓	-	✓
IFND	✗	-	✗
LIAR	✓	-	✗
LIAR-New	✓	✓	✗
MediaEval	-	✗	-
MM-COVID	✗	-	✗
MultiClaim	-	-	✗
NLP4IF-2021	✓	-	✗
PEHEME	-	-	✗
PubHealthTab	✓	-	✗
Rumors	✓	✗	✗
Snopes Fact-news	✓	-	✗
TruthSeeker2023	✗	-	✗
Twitter15	✗	✗	✗
Twitter16	✗	✗	✗
Verite	✓	-	✗
WICO	-	✓	-
X-Fact	✓	✗	✗
BanFakeNews	✗	✗	✗
BenjaminPoliticalNews	✗	-	✗
Celebrity	✗	-	✓
CT-FAN	✗	-	✓
FA-KES	✓	✓	✓
FakeNewsAMT	✓	-	✓
FakeNewsCorpus	-	-	✗
ISOT Fake News	✗	✗	✓
TI-CNN	✗	✗	✓

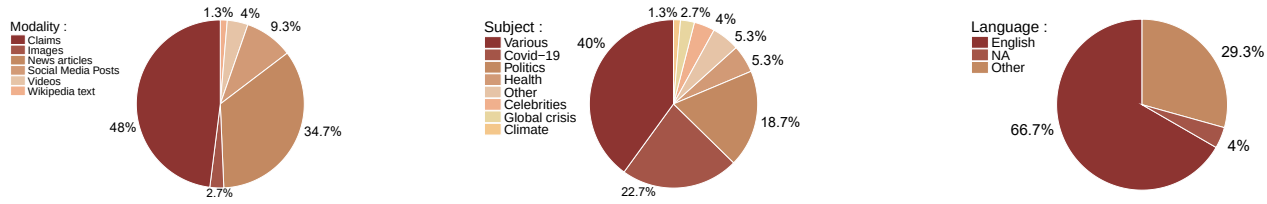


Figure 1: An overview of the modalities, subject counts, and languages of datasets in CDL-MD. For more details on these and other attributes like size, geographic region, and date of the data, please refer to the comprehensive table in the appendix of the extended paper.

2 CDL-MD

We curate a combined collection of 75 datasets, named **CDL-MD** (Complex Data Lab Misinfo Datasets). The collection includes 36 specifically focused on claims and statements (defined here as short texts ranging from one to two sentences), and 9 specifically focused on paragraphs (longer than two sentences). The full collection contains a total of 120,901,495 observations, while the subset that we further analyze includes 1,741,146 observations. These data encompass a wide range of topics, modalities, languages, and other characteristics (partially summarized in Figure 1).

Harmonizing labels across different datasets is a crucial step to ensure comparability of results and robustness of analyses. Since each study employs its own criteria for classifying veracity, in our collection, we use the original labels from the studies to create a more consistent categorical variable across datasets. Specifically, we classify content as true, false, mixed or unknown. Information that is mostly true is classified as *true*, while content that is mostly false is classified as *false*. Finally, ambiguous claims, such as those partially true or false, were classified as *mixed*, while claims that were unproven, unrelated or contained no information about their veracity were coded as *unknown*.

An extensive treatment of the construction process of this collection, including how we identified the datasets, additional descriptive information for each one, expanded discussion of the labeling approach, and other discussion, is included in the appendix of the extended paper.

3 Data Quality

Shortcut learning is a serious barrier to predictive systems working in the real world [10]. In this section, we assess datasets’ potential for teaching algorithms spurious keyword and temporal correlations. These are key considerations when training and evaluating supervised methods. We then assess whether the inputs are making sufficiently complete and unambiguous claims for it to be feasible to predict their veracity at all, which is critical to the evaluation of all types of methods. We provide the automated tools that enable this analysis through our website as the Complex Data Lab Dataset Quality Assessment (CDL-DQA).

3.1 Spurious Keyword Correlations

We first evaluate whether there are certain keywords that over-predict veracity in the datasets. We adapt the approach that [18] used to check for spurious temporal correlations. Specifically, we

trained a random forest classifier with the 40 most frequent words in each dataset, after removing stop-words and excluding datasets containing only tweet IDs (since in that case the claims are inaccessible), as well as those with single veracity labels that include only false statements. This analysis was conducted in two stages: first by incorporating only the labels True and False, and second, by also including the label Mixed. Utilizing scikit-learn, we set a maximum tree depth of 20 and retained the other default settings. We then compare with a baseline of predicting randomly according to the class label distribution and no other information, providing a reference point for assessing the predictive power of the keywords. Thus, keywords in a dataset are significantly predictive if their performance significantly exceeds the baseline. The macro F1 scores for the true and false labels are presented in Figure 2, and Table 6 of the extended paper provides the results that also incorporate mixed labels and baselines.

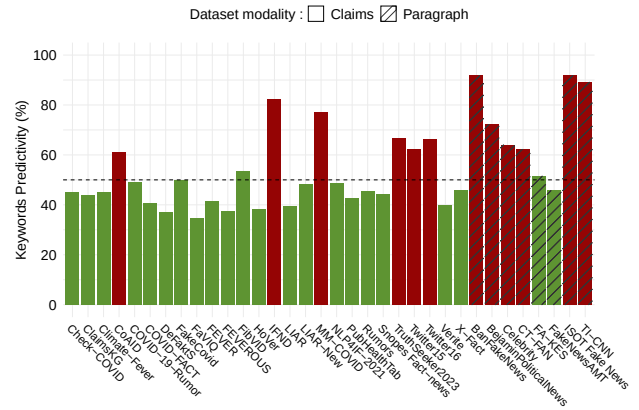


Figure 2: Keywords correlations evaluation. A high predictivity score that far exceeds the 50% baseline, indicated by the dashed line, means that the keywords provide an unrealistically strong prediction. Green bars indicate datasets that pass this check, while red bars represent ones with spurious correlations. All numbers are % macro F1.

We particularly flag six claims datasets for spurious correlations between certain words and labels, as well as six datasets containing paragraphs. Respectively, these are: CoAID, IFND, MM-COVID,

TruthSeeker2023, Twitter15 and Twitter16; and BanFakeNews, BenjaminPoliticalNews, Celebrity, CT-FAN, ISOT Fake News and TI-CNN. For example, consider Truthseeker2023. Nearly all tweets here mentioning politicians are labeled as “false”, with only those containing “Trump” showing more variation (see the appendix of the extended paper). Obviously, in the real world, tweets mentioning politicians are not exclusively false. Thus, models trained on data like Truthseeker2023 risk generalizing inaccurate results, and doing so on topics extremely sensitive to bias like discussion of politicians. Moreover, these findings extend beyond political names. Terms like “michigan”, “vaccines”, “immunity”, “pfizer”, and so on are consistently labeled as false, while words like “marijuana”, “wealth”, “terrorism”, “radical” and others are always associated with the true label. Similar patterns are also observed in the other datasets with highest keyword predictivity. Therefore, we urge caution about training and testing models on these datasets, especially text-focused models.

3.2 Spurious Temporal Correlations

Pelrine et al. [18] highlighted how collecting data of different classes at different times can make temporal information unrealistically predictive. For example, discussion of particular news events can become excessively correlated with veracity labels, leading to artificially inflated performance for classifiers that rely on these events, that will not generalize to the real world where veracity cannot be determined by event or topic alone.

Like in the preceding section, we assess this limitation by training a random forest classifier. As feature, we use either the first three digits of the tweet ID (which contain time information) as in [18] for Tweet datasets, or the date itself for datasets which include it. For the latter, we encode it as the integer number of days since the first date in the dataset. We exclude from this analysis datasets without either form of temporal information.

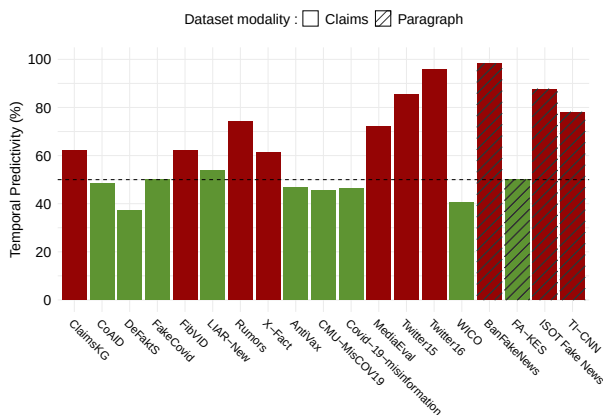


Figure 3: Temporal correlation evaluation. A score that far exceeds the 50% baseline means time—and information correlated with it—is unrealistically predictive. Green bars indicate datasets that pass this check, while red bars represent ones with spurious correlations. All numbers are % macro F1.

Results are shown in Figure 3. We first note that our findings on Twitter15 and Twitter16 are similar to [18], confirming these datasets have extreme issues with spurious correlations in temporal information. They should not be used without carefully and explicitly addressing this limitation. Similarly, the paragraph datasets BanFakeNews, IsotFakeNews, and TI-CNN show high levels of spurious temporal leakage. While not quite as severe, we also see that MediaEval and Rumors also suffer from some significant spurious temporal correlations, and caution is advised. The rest of the datasets have a substantially better temporal balance, with the temporal feature offering little better than random performance. However, we note that only a small fraction of the total datasets include dates, and recommend that future datasets add this important information.

3.3 Feasibility

If a claim is too ambiguous, it may be impossible and meaningless—or even misleading—to assess its veracity, irrespective of the power of one’s assessment system. For example, it is impossible to evaluate the veracity of the claim “The senator said the earth is flat” without knowing which senator is referred to. If a dataset contains too many such examples, it will be problematic to train and evaluate algorithms on it. [19] performed analysis on a limited, manual scale and found problems of this type in the LIAR dataset. To extend this analysis to the field at large, 8 human expert annotators each labeled examples from 29 datasets. We also developed an AI annotator that gives a comparable assessment, providing both additional evidence and scalability.

Our annotators categorized the claim text according to the following schema:

- **Feasible:** The statement provides enough context for an AI to determine its truthfulness with certainty.
- **Feasible with web search:** Some key information is missing, preventing an AI from determining the claim’s truthfulness without retrieving additional data online.
- **Not feasible:** The statement is too vague or incomplete for a web search to provide sufficient evidence for verification.

This was done for 1230 human-annotated claims dataset examples (uniformly distributed for approximately 42 per dataset) with most annotated by at least two annotators, and 8700 AI-annotated claims examples plus 2700 paragraph examples (300 per dataset). Full human annotator instructions and sampling procedures, LLM prompting procedures, and other supplementary information are available in the extended paper.

We first aggregate over datasets, presenting the individual results of our 9 human and AI annotators in Figure 4. Our results show universal agreement that **without an evidence retrieval (search) system, at least half of claims in these 29 datasets cannot be validly assessed for veracity**. This suggests that such methods are often being evaluated on impossible tasks, and there is a severe risk that evaluation comparing such methods is determining not the best generalizable predictor but the best shortcut learner. We recommend that any proposed methods in this category conduct stringent Evaluation Quality Assurance (see Section 5), and suggest that applications requiring even slightly out of distribution predictions should ground them in evidence.

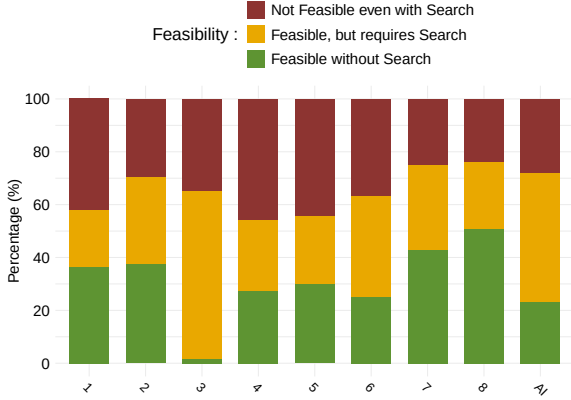


Figure 4: Feasibility assessed by 8 human expert and 1 AI annotators, averaged over all datasets. Without an evidence retrieval system (e.g., web search), most data is not feasible to assess for veracity.

We next focus on systems that *do* have access to retrieval, particularly open web search, and assess which datasets will have strong feasibility and therefore represent strong options for training and evaluation. We propose 75% feasibility as a generous threshold, allowing for a moderate amount of noise in both our assessment and in applications relying on the datasets. We focus on the human annotator average, noting that the AI annotator has decent alignment with human assessment, and advantages in sample size and scalability, but is typically a bit more generous. Regardless, both forms of assessment yield similar conclusions. We note that for some applications, up to a quarter of examples in the evaluation data representing noise may be far too much—for example, comparisons between methods with margins of a couple percentage points. Nonetheless, we see in Figure 5 that most datasets do not even meet this standard.

We next turn to paragraph datasets. We find (Figure 6) that they have a higher feasibility rate than claims datasets. This result makes sense considering that having a greater volume of text allows for more context, which reduces the amount of ambiguity in classifying misinformation. Thus, while it can be more complex to assess veracity with longer inputs, evaluation in this setting may be paradoxically easier. Since recent advances in LLMs (and multimodal systems that can process other types of inputs) provide many options for working with longer data, prioritizing this and other context-rich settings may be beneficial for the field.

We also observe (elaborated in the extended paper) that feasibility has a substantial relationship (e.g., 0.56 Pearson correlation) with the performance of a veracity prediction system *across* different datasets. We note a caveat that it appears very noisy *within* datasets, i.e., feasibility is not a strong predictor for whether an individual example’s veracity will be assessed correctly. This is likely because there are many reasons beyond feasibility that a prediction system can be correct or incorrect, such as data leakage,

Table 2: State-of-the-art GPT-4 baselines, with and without web search.

Dataset	F1 (Search)	F1 (Offline)
Check-COVID	78.8%	85.4%
Climate-Fever	66.9%	65.3%
CoAID	62.0%	60.3%
COVID-19-Rumor	62.8%	65.8%
COVID-Fact	67.5%	67.4%
FakeCovid	50.4%	51.0%
FaVIQ/test	81.5%	80.7%
FEVER/paper_test	88.6%	89.2%
FEVEROUS/validation	65.6%	62.2%
FibVID	67.6%	67.3%
HoVer/validation	68.8%	61.7%
IFND	56.5%	42.0%
LIAR/test	44.8%	50.7%
LIAR-New	69.7%	63.6%
MM-COVID	85.6%	86.5%
PHEME	34.3%	33.4%
PubHealthTab/test	30.8%	49.3%
Rumors	69.5%	80.7%
Snopes Fact-news	90.6%	81.4%
TruthSeeker2023	81.9%	81.0%
Twitter15	57.7%	66.5%
Twitter16	49.2%	55.8%
Verite	63.3%	59.8%
X-Fact/test	55.0%	53.0%

lucky search results or assumptions about context or predictions in general, etc.

Overall, there are 3 main implications. First, there is an urgent need for higher quality claims datasets. Second, supplementary information beyond claim text, such as claim dates, authors, extended paragraphs, or other additional context may help alleviate this problem. However, it is critical to assess how the specific information a predictive method processes impacts the feasibility of the data it is being trained and evaluated on, to confirm that performance margins between methods represent real progress and not progress in predicting noise. Thus, third, we propose that even in the more favorable retrieval-augmented setting, Evaluation Quality Assurance (Section 5) is paramount.

4 Evaluation

4.1 Baseline Performance

We next discuss baselines when using these datasets. Recent works have shown LLMs represent the state-of-the-art for misinformation detection on generic claims [7, 19]. But despite its importance [18], both human and compute time constraints can be a barrier to comparing with such strong baselines.

Thus, we provide two new baselines for future use. We follow the recent method of [24] and use GPT-4-0125 in two ways: directly prompting the LLM for a veracity evaluation, and providing the LLM a web search tool to first collect evidence before forming a final verdict. We note that although these are state-of-the-art

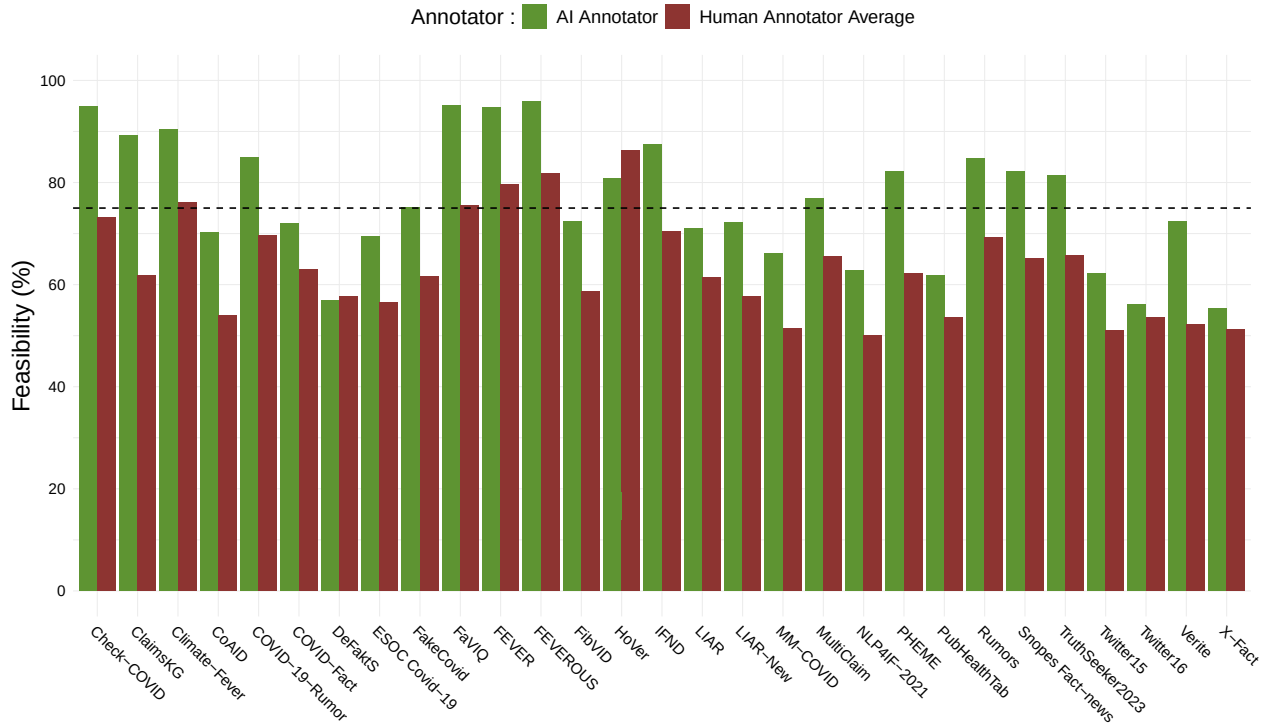


Figure 5: Evaluation of claims dataset feasibility. Even with evidence retrieval, most datasets have a concerning proportion of infeasible data.

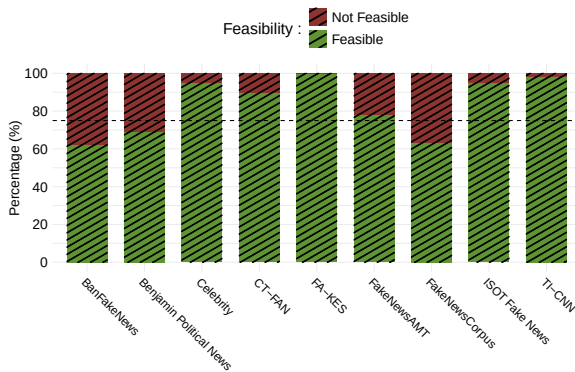


Figure 6: AI annotator evaluation of feasibility over paragraph datasets. These datasets often have higher feasibility proportions than claims datasets.

systems for zero-shot misinformation detection, they should not be regarded as sole or permanent points of comparison. Stronger LLMs and methods could replace them eventually. Nonetheless, they can provide a useful point of comparison for the near future.

We note that 8 datasets are excluded from this baseline: 7 tweet datasets that we were unable to retrieve due to X API limits, and the ESOC Covid-19 dataset because it only has a “refutes” label. Results on all others are provided in Table 2. Notably, because these are zero-shot approaches, they are much less vulnerable to spurious correlations than models trained on each of these datasets, sometimes leading to dramatically lower but more realistic performance compared to alternatives in the literature (e.g., Twitter15 and Twitter16, where temporal classification achieves over 80% F1).

4.2 The Flaw in Current Metrics

When looking at the outputs of the prediction system, we observe cases where the predicted label did not match the ground truth, yet the evidence and reasoning of the system was valid. For instance, in one example on the FEVER dataset, the input claim is “Vietnam is a place” and the prediction said roughly “Vietnam is not just a place, it’s a country!” In another example from LIAR-New, a statement was marked false by PolitiFact because it was in the context of a fake video, but the statement itself did not mention the video and in isolation would be true. In cases like these, a simple binary or categorical label cannot provide an informative evaluation.

To determine the prevalence of this phenomenon, two human expert annotators evaluated chain-of-thought rationales from the web-search enabled baseline prediction system [24].

Table 3: Agreed-upon manual annotations. Many predictions marked invalid by simple comparison with the categorical labels are actually valid. Standard evaluations are likely to systematically under-evaluate the performance of generative systems.

AI rationale is:	Label $\neq$ Prediction		Label = Prediction	
	Valid	Invalid	Valid	Invalid
LIAR-New	55/100	30/100	76/100	1/100
FEVER	38/100	34/100	–	–
MM-COVID	39/70	3/70	89/100	0/100

We observe a consistently high false-incorrect rate (first column of Table 3) and a generally low false-correct rate (fourth column of Table 3). Therefore, when benchmarking generative AI misinformation detection systems using categorical labels, the predictive accuracy and similar metrics reflect a reasonable lower bound on the performance—but a terrible upper one, that marks many valid responses incorrect. We also observe that there is a large amount of ambiguity and room for interpretation in the examples that are being marked wrong by categorical label in these three datasets. Hence:

- (1) Categorical metrics cannot be used alone to compare generative and non-generative systems. Although multiple recent works (e.g., Chen and Shu [7], Pelrine et al. [19], Wei et al. [26], Yu et al. [28]) have highlighted the effectiveness of recent LLMs for misinformation detection, their comparisons with prior approaches may even still be underestimating the dominance of LLMs in this domain.
- (2) Generative systems need many, repeated, and large-margin measurements if the categorical lower bound alone is to form meaningful comparisons between them.
- (3) There is an urgent need for better datasets and better evaluation procedures in this domain that are suitable for the generative AI era.

Although to our knowledge not addressed in the context of misinformation detection on claims, challenges in evaluations of generative AI have been broadly documented in other fields [2, 5, 14, 15]. A common approach aimed at solving this is having an LLM assess the full prediction rationales [4, 23], with Sun et al. [22] noting that LLM-powered evaluation can produce more consistent preference signals than human annotators. In general, using LLMs for evaluation enables one to leverage much richer signals than simple categorical predictions and labels, while avoiding reliance on often inaccessible human evaluators.

As an initial step towards higher-fidelity evaluation, we constructed an evaluator based on contradictions between the explanation generated by a predictive system, and a fact-checking article. In particular, we provide GPT-4-0409 both *the prediction* and *the fact-checking article*, and ask it to score contradictions from 0 (no contradiction) to 10. The exact prompt and other implementation details are provided in the extended paper. We chose a score-based approach to avoid forcing a potentially misleading binary in cases where there is a partial contradiction. With this approach, good

predictions should have low contradiction against a high quality, professional fact-checking article. We also tested binary and trinary versions of this prompt, which yielded nearly identical results.

To evaluate this evaluator, we set the oracle-optimal threshold of 3 or less (low contradiction) to indicate a prediction that the rationale is not wrong, and 4 or more indicating one that the rationale is wrong. We find that the contradiction score evaluator agrees 68% of the time with the human labels of valid and invalid rationales (reported in Table 3) on the LIAR-New dataset. This is higher than the 60% original human inter-annotator agreement on these rationales (before disagreement resolution described in the extended paper), and suggests the method extracts a meaningful but not definitive evaluation signal. We also note, though, that there is more to high quality misinformation detection than just a lack of contradiction. Therefore, we do not suggest using this tool as a primary evaluator. But we provide these results, along with all the data—inputs, predictions, manual labels, fact-checking articles, and evaluator outputs—on our GitHub, as a potential springboard for stronger evaluation methods in future research.

5 Evaluation Quality Assurance

We have highlighted multiple ways datasets and evaluation procedures could deliver results that will fail to translate to the real world. While the information here directly reveals some pitfalls, we suspect that future research—with new datasets, new methods, and new needs—will encounter similar problems. To improve the validity of conclusions in our field in a systematic and future-proof way, we propose *Evaluation Quality Assurance (EQA)* as a key component of an improved research methodology.

We draw inspiration from the common practices for systematically and transparently documenting dataset publications, such as “datasheets for datasets” [9], and suggest that similar attention should be given to the *usage* of datasets. However, it is important to ensure data and evaluation quality not just as a lengthy post-hoc appendix, but as an integral part of the research process and a key foundation of every paper’s core arguments. Therefore, we propose including EQA as a scientifically essential step within the methodology. This means augmenting the standard *descriptive* discussion of data used with a *critical* analysis of it, explaining what steps and quality assessments were conducted to assure that the data is suitable to prove the experimental conclusions. We suggest that dedicating even just one paragraph to this, rather than taking quality of “standard” datasets for granted, could dramatically improve the validity of our empirical results and the rate of real progress in the field. Conversely, current practices risk misleading evaluations, generalization failures, and immeasurable time and effort dedicated to measurements of noise.

We make two key recommendations for Evaluation Quality Assurance.

- **Quality Assessment:** analyze the data and evaluation processes themselves, not just their results, to assure they are suitable to test performance and generalization of a method. One basic analysis, that can be nearly universally recommended, is to qualitatively inspect at least 25 random examples—input, output, and ground truth—where the method made correct predictions, and 25 where it made incorrect ones. This data

can be shared for reproducibility, and provides a basic sanity check. This simple step could already have a high impact; for example, the feasibility and categorical metrics issues we have reported would be detected by such a sanity check in the majority of the datasets that we examined. Furthermore, we also recommend papers perform quantitative assessments of their data and evaluation, such as (but not limited to) the keyword, temporal, and feasibility assessments discussed in prior sections and available in our **CDL-DQA** toolkit.

- **Assurance Limitations:** report **limitations in evaluation quality assurance (EQA)**, not only in terms of dataset size or experimental scope but also in terms of quality issues that the EQA could not rule out. Key to this is considering how a method would be applied in the real world—and how the evaluation might not be fully capturing the real-world conditions. If, for instance, a method is intended to handle diverse contexts or domains of misinformation, but was only tested on data from a single context (e.g., one country) or domain (e.g., politics), this should be stated as a limitation so that future research can address it. Explicitly acknowledging such gaps strengthens scientific rigor and ensures that claims are appropriately supported by the evidence provided.

In this way, we treat assuring evaluation quality as a crucial step for validating the research process. For authors, this will help reinforce the demonstrated rigour of the experiments and the validity of proposed methods. It will also prevent the rejection of strong methods due to unrealistically high baseline settings caused by spurious correlations or other evaluation issues. For reviewers, this will produce more grounding (or expose the lack thereof) for the experiments, as well as a framework for recommendations when evaluation is insufficiently justified and needs better validation. Thus, we propose that adopting EQA can efficiently and substantively mitigate propagating and compounding errors, that arise in this domain from current practices of copy-pasting evaluation from past literature.

6 Related Work

In recent years, the scientific community has devoted considerable research to curbing misinformation, defined broadly as “false or misleading information” [13]. Initial surveys, such as Shu et al. [21], provided foundational frameworks and overviews of detection techniques from data mining perspectives, while subsequent studies by Oshikawa et al. [16] and Zhou and Zafarani [29] emphasized NLP methods and interdisciplinary approaches. Despite these contributions, most existing surveys [6, 11] largely overlook detailed analyses of misinformation datasets, offering superficial coverage that neglects dataset-specific qualities essential for accurate model development and validation. Recent surveys partially address this gap by categorizing datasets quantitatively [3, 8, 17, 20], yet typically lack in-depth analysis. Other works, including Abdali [1], Pelrine et al. [18], and Wu and Hooi [27], provide deeper insights into dataset biases and limitations but remain limited by analyzing only a small subset of datasets. To address these limitations, our work evaluated a broad selection of misinformation datasets, assessed in depth how issues in data and evaluation could lead to

misleading conclusions, and provided tools and recommendations for future improvements in the field.

7 Limitations

First, we note that while to our knowledge this represents the largest and most comprehensive survey of datasets in this domain, there are certainly many other datasets in existence and it is probable that some were not included. There is also a steady stream of new datasets being created every year. In the near future, we plan to collect external feedback and update our survey to maintain and expand the comprehensiveness of our study.

We also note that our unified label schema simplifies some labels that might have meaningful information, for example, gradations of veracity instead of binary true/false. Some granularity has been traded for the ability to establish a unified schema across all the claims datasets. When using these datasets, we advise careful consideration of the optimal labels to apply.

As discussed previously, additional work is needed in evaluation, both to confirm that the observed validity issues with metrics like accuracy are widespread (as we hypothesize) and to create strong, thoroughly tested alternatives. We also note that the baselines we have provided use old evaluation procedures on LLM-based predictors. This can be flawed both for the reasons discussed in Section 4.2, and potentially also because a substantial proportion of the data could be within the LLM training data. Pelrine et al. [19] indicates LLM-based methods offer the strongest performance even beyond their knowledge cutoffs, and using web search to actually provide evidence can mitigate this to some degree. But nonetheless, these baselines should be viewed carefully and with due attention to both their strengths and limitations, and future work to establish more universal baselines—as well as datasets and evaluation methods that enable them—would be very valuable.

Lastly, although we discussed multiple key dimensions of misinformation detection datasets, a favorable assessment in these dimensions does not guarantee a dataset suitable for any application. Other types of limitations in the data could still lead to spurious shortcuts instead of generalizable predictions and evaluation. Our proposed EQA procedure can help detect this with respect to particular methods and applications, but we also encourage future work to identify additional limiting factors in evaluation in this domain, and to solve them.

8 Conclusion

High quality data and evaluation are essential for realistic results and rapid progress in the field. In this work, we have provided a guide to misinformation detection datasets aiming at both quantity and quality. We also highlighted limitations of existing datasets and evaluation approaches, which may have spurious correlations, infeasible examples, and misleading results. We hope that this work can provide a roadmap to better grounding for future predictive methods research that needs to select datasets and evaluation approaches. Meanwhile, we also hope that this guide will provide foundational understanding and a call to action to build new and better datasets and evaluation procedures.

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References

- [1] Sara Abdali. 2024. Multi-modal misinformation detection: Approaches, challenges and opportunities. *arXiv preprint arXiv:2203.13883* (2024).
- [2] Kabir Ahuja, Harshita Diddee, Rishav Hada, Millicent Ochieng, Krithika Ramesh, Prachi Jain, Akshay Nambi, Tanuja Ganu, Sameer Segal, Mohamed Ahmed, et al. 2023. MEGA: Multilingual Evaluation of Generative AI. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. 4232–4267.
- [3] Ihsan Ali, Mohamad Nizam Bin Ayub, Palaiahnakote Shivakumara, and Nurul Fazmidar Binti Mohd Noor. 2022. Fake news detection techniques on social media: A survey. *Wireless Communications and Mobile Computing* 2022, 1 (2022), 6072084.
- [4] Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, et al. 2022. Constitutional ai: Harmlessness from ai feedback. *arXiv preprint arXiv:2212.08073* (2022).
- [5] Rahul C Basole and Timothy Major. 2024. Generative AI for Visualization: Opportunities and Challenges. *IEEE Computer Graphics and Applications* 44, 2 (2024), 55–64.
- [6] Alessandro Bondielli and Francesco Marcelloni. 2019. A survey on fake news and rumour detection techniques. *Information sciences* 497 (2019), 38–55.
- [7] Canyu Chen and Kai Shu. 2023. Combating misinformation in the age of llms: Opportunities and challenges. *arXiv preprint arXiv:2311.05656* (2023).
- [8] Arianna D’Ulizia, Maria Chiara Caschera, Fernando Ferri, and Patrizia Grifoni. 2021. Fake news detection: a survey of evaluation datasets. *PeerJ Computer Science* 7 (2021), e518.
- [9] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé Iii, and Kate Crawford. 2021. Datasheets for datasets. *Commun. ACM* 64, 12 (2021), 86–92.
- [10] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A Wichmann. 2020. Shortcut learning in deep neural networks. *Nature Machine Intelligence* 2, 11 (2020), 665–673.
- [11] Georgios Gravanis, Athena Vakali, Konstantinos Diamantaras, and Panagiotis Karadais. 2019. Behind the cues: A benchmarking study for fake news detection. *Expert Systems with Applications* 128 (2019), 201–213.
- [12] Suhaib Kh Hamed, Mohd Juzaidin Ab Aziz, and Mohd Ridzwan Yaakub. 2023. A review of fake news detection approaches: A critical analysis of relevant studies and highlighting key challenges associated with the dataset, feature representation, and data fusion. *Heliyon* (2023).
- [13] David MJ Lazer, Matthew A Baum, Yochai Benkler, Adam J Berinsky, Kelly M Greenhill, Filippo Menczer, Miriam J Metzger, Brendan Nyhan, Gordon Pennycook, David Rothschild, et al. 2018. The science of fake news. *Science* 359, 6380 (2018), 1094–1096.
- [14] Timothy R McIntosh, Teo Susnjak, Tong Liu, Paul Watters, and Malka N Halgamuge. 2024. Inadequacies of large language model benchmarks in the era of generative artificial intelligence. *arXiv preprint arXiv:2402.09880* (2024).
- [15] Rosario Michel-Villarreal, Eliseo Vilalta-Perdomo, David Ernesto Salinas-Navarro, Ricardo Thierry-Aguilera, and Flor Silvestre Gerardou. 2023. Challenges and opportunities of generative AI for higher education as explained by ChatGPT. *Education Sciences* 13, 9 (2023), 856.
- [16] Ray Oshikawa, Jing Qian, and William Yang Wang. 2018. A survey on natural language processing for fake news detection. *arXiv preprint arXiv:1811.00770* (2018).
- [17] Debasish Patra, Biswapati Jana, Sourav Mandal, and Arif Ahamed Sekh. 2022. Understanding fake news detection on social media: A survey on methodologies and datasets. In *International Symposium on Artificial Intelligence*. Springer, 226–242.
- [18] Kellin Pelrine, Jacob Danovitch, and Reihaneh Rabbany. 2021. The surprising performance of simple baselines for misinformation detection. In *Proceedings of the Web Conference 2021*. 3432–3441.
- [19] Kellin Pelrine, Anne Imouza, Camille Thibault, Meilina Reksoprodjo, Caleb Gupta, Joel Christoph, Jean-François Godbout, and Reihaneh Rabbany. 2023. Towards Reliable Misinformation Mitigation: Generalization, Uncertainty, and GPT-4. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. 6399–6429.
- [20] Karishma Sharma, Feng Qian, He Jiang, Natali Ruchansky, Ming Zhang, and Yan Liu. 2019. Combating fake news: A survey on identification and mitigation techniques. *ACM Transactions on Intelligent Systems and Technology (TIST)* 10, 3 (2019), 1–42.
- [21] Kai Shu, Amy Sliva, Suhang Wang, Jiliang Tang, and Huan Liu. 2017. Fake news detection on social media: A data mining perspective. *ACM SIGKDD explorations newsletter* 19, 1 (2017), 22–36.
- [22] Zhiqing Sun, Yikang Shen, Hongxin Zhang, Qinhong Zhou, Zhenfang Chen, David Cox, Yiming Yang, and Chuang Gan. 2023. SALMON: Self-Alignment with Principle-Following Reward Models. *arXiv:2310.05910* [cs.LG]
- [23] Zhiqing Sun, Yikang Shen, Qinhong Zhou, Hongxin Zhang, Zhenfang Chen, David Cox, Yiming Yang, and Chuang Gan. 2023. Principle-Driven Self-Alignment of Language Models from Scratch with Minimal Human Supervision. In *Thirty-seventh Conference on Neural Information Processing Systems*. <https://openreview.net/forum?id=p40XRfBX96>
- [24] Jacob-Junqi Tian, Hao Yu, Yury Orlovskiy, Tyler Vergho, Mauricio Rivera, Mayank Goel, Zachary Yang, Jean-François Godbout, Reihaneh Rabbany, and Kellin Pelrine. 2024. Web Retrieval Agents for Evidence-Based Misinformation Detection. In *First Conference on Language Modeling*.
- [25] Simon Torkington. 2024. AI-driven disinformation and global risks. *World Economic Forum Agenda* (2024). <https://www.weforum.org/agenda/2024/01/ai-disinformation-global-risks/> Accessed: 2024-09-20.
- [26] Jerry Wei, Chengrun Yang, Xinying Song, Yifeng Lu, Nathan Hu, Dustin Tran, Daiyi Peng, Ruibo Liu, Da Huang, Cosmo Du, et al. 2024. Long-form factuality in large language models. *arXiv preprint arXiv:2403.18802* (2024).
- [27] Jiaying Wu and Bryan Hooi. 2022. Probing spurious correlations in popular event-based rumor detection benchmarks. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*. Springer, 274–290.
- [28] Hao Yu, Zachary Yang, Kellin Pelrine, Jean Francois Godbout, and Reihaneh Rabbany. 2023. Open, Closed, or Small Language Models for Text Classification? *arXiv preprint arXiv:2308.10092* (2023).
- [29] Xinyi Zhou and Reza Zafarani. 2020. A Survey of Fake News: Fundamental Theories, Detection Methods, and Opportunities. *ACM Comput. Surv.* 53, 5, Article 109 (sep 2020), 40 pages. doi:10.1145/3395046
- [30] Arkaitz Zubiaga, Geraldine Wong Sak Hoi, Maria Liakata, and Rob Procter. 2016. PHEME dataset of rumours and non-rumours. (10 2016). doi:10.6084/m9.figshare.4010619.v1